

Will changes to the shoreline vegetation due to the new reservoir for the Site C Dam impact the shoreline stability and health of the riparian zone along the Peace River in British Columbia between the Moberly River and Cache Creek

INTRODUCTION – *Reservoirs created in existing rivers cause long-term changes to the shoreline vegetation, riparian zone and slope stability*

Reservoir creation through river impoundment causes strong changes to flow regime and biotic environments, specifically effecting species tolerant of niche conditions (Meißner et al., 2018; Zwahlen, 2022). The resulting transition from a river to a lake-like reservoir causes significant vegetation shifts (Zwahlen, 2022), including species elimination due to anoxic waterlogged soils (Nilsson & Berggren, 2000) and the emergence of vegetation that thrives in fluctuating water levels (Zwahlen, 2022). On the Nile River, the Aswan High Dam reservoir increased water levels which altered the flow regime and resulted in the growth of new plant species and vegetation spatial distribution (Springuel et al., 1991). In many arid US States, reservoirs increased the river water levels and created a suitable environment for the invasion of exotic species, decreasing riparian diversity (New & Xie, 2008). In contrast, the steeper slopes of larger reservoirs on the Missouri River have resulted in a loss of pre-reservoir vegetation and no new riparian zone formation, triggering increased reservoir sedimentation and unstable shorelines (Johnson, 2002).

The 2013 Environmental Impact Statement and 2014 Report of the Joint Review Panel completed for the Site C Dam and Generating Station situated on the Peace River, BC predicted that the reservoir would have numerous environmental consequences to the surrounding ecosystems and their economic values (Report of the Joint Review Panel, 2014). They highlighted that the movement from a river to reservoir habitat will result in notable changes to the shoreline, including the replacement of existing lotic vegetation with littoral vegetation (Report of the Joint Review Panel, 2014) and a decrease in riparian zone diversity (Hydro & Street, 2012), increasing reservoir bank slope failure risk and long-term stability (Chen et al., 2023; Hydro & Street, 2012).

In late 2024 the Site C Dam's reservoir was filled completely (Filling the Reservoir | Site C, n.d.). It spans a length 83 km of varying depth with an estimated surface area of 9330 hectares between Fort St. John and Hudson's Hope (Filling the Reservoir | Site C, n.d.; Report of the Joint Review Panel, 2014). Early reports have indicated that permanent changes along the shoreline have already been observed (Filling the Reservoir | Site C, n.d.). With this study I aim to identify how the shoreline vegetation changes upon the filling of this new reservoir influence fluctuations in riparian zone health and degradation of slope stability.

METHODS – *An analysis using a combination of field and remote sensing data over a series of years can aide in predictive mapping and modelling of the impacted area*

This study will be conducted between Peace River tributaries Moberly River and Cache Creek. I will complete a twenty-year time series analysis at intervals of 2015 (pre construction), 2025 (one year post reservoir fill), 2030 (five years post reservoir fill) and 2035 (twenty years after initial construction). Landsat 8/9 imagery will be acquired to conduct a supervised classification to identify vegetation composition and compute NDVI to evaluate the riparian zone. A DEM will be used to derive slope on the north and south river banks. Field data collected over 10 permanent plots in 2015 and 2025 on both river banks will validate the remote sensing analysis. I will create a predictive model to determine the relationship between reservoir age, shoreline vegetation composition, riparian zone health and shoreline stability. The trends identified between 2015 and

2025 in correlation with historical information and prior knowledge will be used to calculate the predicted metrics and maps.

EXPECTED RESULTS AND SIGNIFICANCE – *The Site C Reservoir causes large-scale changes within the existing environment as it transitions from a river to reservoir ecosystem*

I expect the environmental changes will be gradual, never returning to pre-reservoir conditions, with a noticeable shift towards littoral vegetation along the shoreline (Report of the Joint Review Panel, 2014) one year post reservoir fill and completely 10 years later (Figure 1). The new riparian zone boundary will encompass a previous terrestrial ecosystem (Hydro & Street, 2012) and I expect a noticeable decrease in riparian health one year post reservoir fill and an increase once the new reservoir ecosystem approaches equilibrium (Figure 2). I expect the reservoir bank slope stability will remain low one year post reservoir fill followed by years of instability before the slope degree and erosion rates balance (Figure 3).

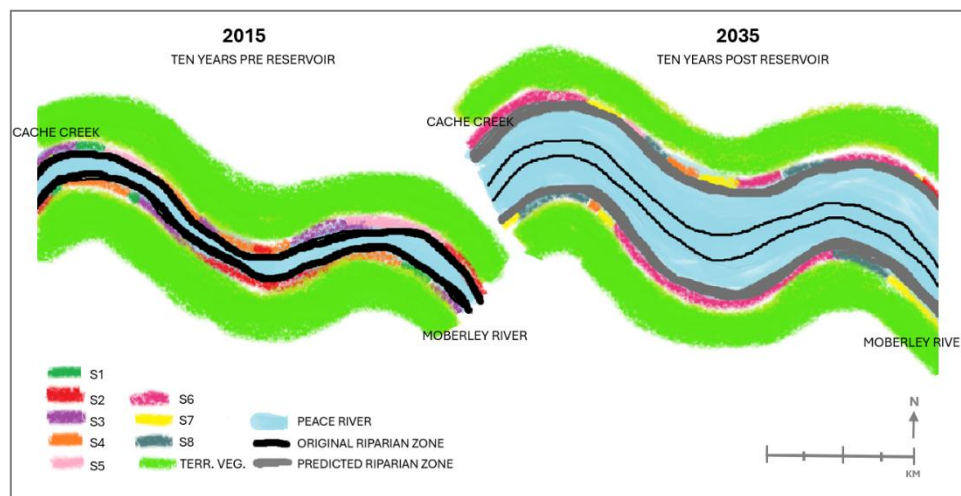


Figure 1: Predicted changes to vegetation composition and riparian zone location. In 2015 the landscape is dominated by the S1 – S5 vegetation and is surrounded by significant terrestrial vegetation. In 2035 the species composition has shifted to include S6 – S7 vegetation which have outcompeted S1, S2, S3 (no longer present on the landscape). The riparian zone now sits in what was once an area dominated by terrestrial vegetation.

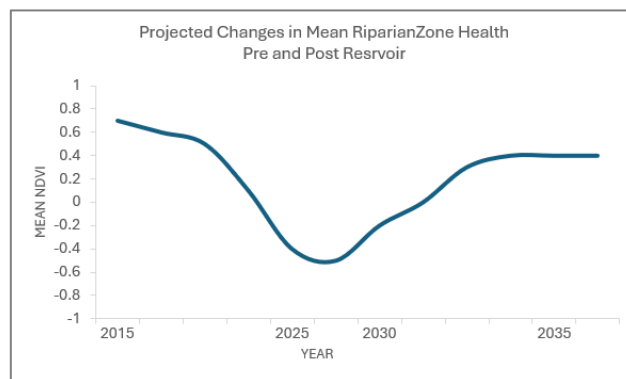


Figure 2: Riparian zone health calculated using NDVI displaying a negative trend from 2015 to 2025. It is predicted that NDVI will slowly increase as the new reservoir ecosystem gets established.

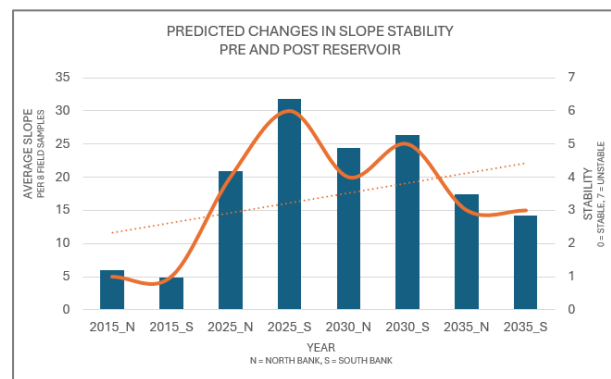


Figure 3: Field data from 2015 and 2025 predicts a decreasing stability rating with average slope values changing as the reservoir alters the vegetation composition on the shorelines.

These results will confirm the severe ecological impacts reservoirs have on the environments in which they are created (Meißner et al., 2018). The predictive data will inform interested stakeholders of ecosystem changes and subsequent long-term environmental, economic and social impacts. These results can be referenced when deciding on future hydroelectric dam projects as they pose the question of: although hydroelectric dams provide clean energy in large quantities, is this positive enough to outweigh the resulting consequences to the stability of a river ecosystem.

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All information in the methods section is my own knowledge gained through educational experience. No external sources were consulted regarding the processes.

No AI Used.